Video Script: Verifying Age Online Without Sacrificing Privacy

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Opening

[Visual: A typical website popup asking "Are you over 18?" with Yes and No buttons.]

Hi everyone.

Let me start with a question.

Have you ever visited a website that asks, *"Are you over 18?"*

And then gives you two buttons:

Yes.

Or...

No.

Now, unless someone is having a remarkably honest day, that isn't exactly the world's most secure verification system.

But that simple question points to a much bigger challenge:

How do we verify someone's age online without turning the internet into an identity-checking machine?

That's the question I explored in my research.

Why Age Verification Matters

[Visual: Icons representing online platforms, government regulation, and protected content.]

As governments and online platforms try to protect minors from age-restricted content, age verification is becoming increasingly common.

The goal sounds simple enough:

Make sure people are old enough to access certain services.

The problem is that proving your age online is surprisingly difficult.

Ideally, a system should be:

- Secure

- Accurate
- Easy to use
- and Respectful of privacy

Unfortunately, those goals don't always get along.

Current Approaches

1. Self-Declaration

[Visual: User typing a fake date of birth into a form.]

The simplest method is self-declaration.

You type in your date of birth and the website trusts you.

It's quick.

It's easy.

It's also about as effective as bringing a butter knife to a rocket launch.

2. Identity Documents

[Visual: Passport and ID card appearing on screen.]

Another option is uploading official documents, like a passport or identity card.

This provides much stronger proof of age.

But now websites are handling highly sensitive personal information.

And that raises a new question:

Do all these websites really need to know who you are?

3. AI Age Estimation

[Visual: Selfie camera interface estimating a person's age.]

More recently, some services have started using artificial intelligence to estimate age from a selfie.

Which sounds futuristic.

But it also raises concerns about accuracy, bias, and what happens to all that biometric data.

4. Digital Identity

[Visual: Digital wallet showing a simple "18+ Verified" credential.]

One approach attracting a lot of attention in Europe is digital identity.

Instead of repeatedly sharing your passport with different websites, you could store verified credentials digitally and simply prove that you're over a certain age.

Not your name.

Not your address.

Just the fact that you're old enough.

Think of it like showing a bouncer a special cryptographicly verifiable card that says "18+" without revealing everything else about you.

The European Age Verification App

[Visual: Simple animation of a user presenting a digital credential to a website.]

That's the idea behind the European Age Verification App, which became the main focus of my research.

The concept has some clear privacy advantages.

But it also creates new security challenges.

If these digital credentials become valuable, they also become attractive targets for theft, fraud, and misuse.

Lessons from Banking

[Visual: Bank verification process showing multiple security checks.]

To understand how strong identity systems can work in practice, I looked at another industry that's been dealing with verification for years:

Banking.

Banks don't simply take your word for it when you open an account.

They combine multiple checks, including:

- Document verification
- Biometric comparison
- Authenticity checks
- Validation against trusted records

These systems have evolved over decades and provide useful lessons for age verification.

One important lesson is that security and trust rarely come from a single check.

They come from combining multiple layers of assurance.

My Proposed Alternative

[Visual: Diagram showing Website → Government Verification Service → Age Confirmation.]

Based on my findings, I believe privacy-preserving age verification is possible, but there is still room for improvement.

In particular, I explored an alternative approach.

Instead of requiring users to install a dedicated app, age verification could be handled through a secure government-operated website.

When a website needs proof of age, you would be redirected to this service, authenticate using modern technologies like passkeys, and receive confirmation that you're old enough.

Importantly, the system would only share the information that's actually needed.

The website learns you're over 18.

Nothing more.

And the best part, the verification service wouldn't know which website you're visiting either.

Conclusion

[Visual: Balanced scale showing Privacy on one side and Online Safety on the other.]

The result is a balance between two legitimate goals:

- Protecting minors from age-restricted content.
- Protecting everyone's privacy.

Because if age verification becomes a normal part of the internet, we shouldn't have to choose between safety and privacy.

The challenge is designing systems that deliver both.

Thank you for listening.